

# Project Design Document

**Project title: SCRIPT-CONTROLLED ACCESS CONTROL LIST (ACL) TO RESTRICT RECORD ACCESS BASED ON FIELD VALUE IN SERVICENOW**

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## ABSTRACT

The project design defines the structure of a ServiceNow-based access control system for protecting student/institution records. The system uses users, custom roles, ACLs, and a field-based condition to control access to records.

The design provides separate permissions for Read, Write, Create, and Delete operations. The Read operation is further restricted based on the **Branch = EEE** condition.

## SYSTEM DESIGN

The system consists of the following major components:

- **User:** EEE User
- **Roles:** bb1, bb2, bb3, bb4
- **Table:** Institution/Student Details
- **ACLs:** Read, Write, Create, Delete
- **Field Condition:** Branch = EEE
- **Script:** Used in the Read ACL for role verification

### Basic System Flow

**User → Role Verification → ACL → Condition/Script → Allow or Deny Access**

## TABLE DESIGN:

The system uses an Institution/Student Details table with the following fields:

| Field Name          | Purpose                       |
|---------------------|-------------------------------|
| Student Roll Number | Auto-generated student number |
| Student Name        | Stores student name           |
| Faculty Name        | Stores faculty information    |
| Branch              | Stores branch information     |
| Phone Number        | Stores contact number         |
| Description         | Stores additional details     |

The **Branch** field is important because it is used for the Read access restriction.

## ROLE DESIGN

| Role | Permission |
|------|------------|
| bb1  | Read       |
| bb2  | Write      |
| bb3  | Create     |
| bb4  | Delete     |

## ACL DESIGN

The ACL structure is:

### ACL Operation Role Condition Script

Read ACL Read bb1 Branch = EEE Yes

Write ACL Write bb2 None No

Create ACL Create bb4 None No

Delete ACL Delete bb4 None No

The Read ACL uses a script to verify the user's role, while the other ACLs use role-based access without scripting.

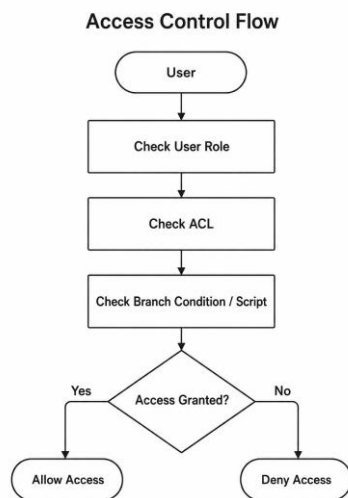
## SECURITY DESIGN

The system follows role-based and field-based access control.

- bb1 controls Read access.
- bb2 controls Write access.
- bb3 controls Create
- bb4 Delete access.
- Read access is restricted using **Branch = EEE**.
- Users without the required role are denied the corresponding operation.

This design helps protect student information from unauthorized access or modification.

## ACCESS CONTROL FLOW



For example:

**EEE User → bb1 → Branch = EEE → Read Access**

## EXPECTED OUTCOME:

The designed system should provide controlled access to student records based on user roles and the required branch condition.

The final access structure is:

**bb1 → Read**

**bb2 → Write**

**bb3 → Create**

**bb4 ---→ Delete**

**Branch = EEE → Read restriction**

## CONCLUSION

The project design provides a structured approach for implementing script-controlled ACLs in ServiceNow. The use of separate roles for Read, Write, Create, and Delete operations, along with the **Branch = EEE** condition, helps control access to student records. The design provides a simple and secure structure for preventing unauthorized access and protecting institutional information.